

WILLIAM NIXON

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🌐 [williamnixon20](https://williamnixon20.github.io)

🌐 [Website](#)

RESEARCH PROFILE

Computer Science Ph.D. student at the University of Chicago researching efficient systems for caching, storage, memory management, and LLM serving. Seeking systems and infrastructure internships for **Summer 2027**.

EDUCATION

University of Chicago

Ph.D. in Computer Science; GPA: 4.00

Advisors: Prof. Haryadi S. Gunawi and Prof. Juncheng Yang (Harvard)

Sep 2025 – Present

Chicago, IL

Bandung Institute of Technology

B.S. in Computer Science; GPA: 3.95

TOEFL: 115 (Reading: 30, Listening: 29, Speaking: 27, Writing: 29)

Jul 2021 – Jul 2025

Bandung, Indonesia

PUBLICATIONS

*Presenting author

- [p1] **Learning-Augmented Heuristics: Simple yet Smart, Robust and Interpretable Cache Eviction** OSDI'26
Haocheng Xia, William Nixon*, Bintang Dwi Marthen, Pranav Bhandari, Juncheng Yang.
Proceedings of the 20th USENIX Symposium on Operating Systems Design and Implementation, 2026.
- [p2] **GLANCED-IO: Taming I/O Optimization for Deep Learning at Scale** HPDC'26
Ray A. O. Sinurat, William Nixon, Philip Carns, ... Haryadi S. Gunawi, Hariharan Devarajan.
Proceedings of the 35th ACM International Symposium on High-Performance Parallel and Distributed Computing, 2026.
- [p3] **HORATIO: Bridging Management and Analysis of Traces at Scale** SSDBM'26
Ray A. O. Sinurat, William Nixon, Haryadi S. Gunawi, Nikoli Dryden, Hariharan Devarajan.
Proceedings of the 38th International Conference on Scientific and Statistical Database Management, 2026.
- [p4] **Crowdsourc, Crawl, or Generate? Creating SEA-VL, a Multicultural Vision-Language Dataset for Southeast Asia** ACL'25
Samuel Cahyawijaya, Holy Lovenia, Joel Ruben Antony Moniz, ... William Nixon, ... Peerat Limkonchotiwat.
Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), 2025.
- [p5] **SEACrowd: A Multilingual Multimodal Data Hub and Benchmark Suite for Southeast Asian Languages** EMNLP'24
Holy Lovenia, Rahmad Mahendra, Salsabil Maulana Akbar, ... William Nixon, ... Samuel Cahyawijaya.
Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing.
- [p6] **From Requests to Sessions: A Large-Scale Characterization of Human-Driven Agentic Workloads** Preprint
William Nixon, Muxin Tian, Yunjia Zheng, Haryadi S. Gunawi, Juncheng Yang.
In submission, 2026.
- [p7] **A Year in LLM Serving: Workload Evolution, Caching and Load-Balancing** Preprint
William Nixon, Jon Durbin, Florian Standhartinger, Haryadi S. Gunawi, Juncheng Yang.
In submission, 2026. arxiv.org/abs/2608.13573
- [p8] **Demystifying and Optimizing Slab Rebalance in Memory Caches** Preprint
Hongshu Yan, Yunjia Zheng, William Nixon, Yazhuo Zhang, Pranav Bhandari, Haryadi S. Gunawi, Ana Klimovic, Juncheng Yang.
In submission, 2026.

RESEARCH EXPERIENCE

Caching and Memory Management

Sep 2025 – Present

- **Learning-Augmented Heuristics, LAH (OSDI'26):** Uses ML to make simple cache heuristics adaptive by automatically tuning their parameters, outperforming more complex ML-based cache policies with negligible runtime overhead; improves cache efficiency by 26% over S3-FIFO and 8% over prior state of the art.
- **Adaptive Slab Rebalancing (Preprint):** Improves memory allocation in slab-based caches by stabilizing noisy rebalance signals and preventing unnecessary slab movements; production deployment reduces DRAM misses by 20% and CPU utilization by up to 5%.
 - Implemented S4-FIFO and Adaptive-Marginal-Hits in C++/CacheLib and worked with Meta engineers to deploy and evaluate them in production.

LLM Serving and Agentic Workloads

Jul 2026 – Present

- **Agentic Workload Analysis (Preprint):** Analyzes 1.2M production requests, 205B input tokens, and 12K coding-agent sessions across 13 agent frameworks; characterizes tool execution, context growth, and KV-cache reuse, and releases the largest production agentic trace with tool- and block-level information to date.
- **A Year in LLM Serving (Preprint):** Analyzes and releases a year-long production trace of 6.1B LLM requests across 9K+ models, the largest public LLM serving dataset to date, enabling large-scale studies of workload evolution, prefix caching, and request routing.
 - Led both projects: built the analysis pipeline for TBs of production requests and the event-driven simulator for cache eviction and request routing, and maintain the open-source trace and benchmark repositories for release.

Deep Learning I/O and Trace Analysis

Mar 2025 – Jul 2026

- **GLANCED-IO (HPDC'26):** Optimizes I/O bottlenecks in large-scale weather-forecasting workloads by tuning I/O and data configurations through GPU-free proxy applications, improving training performance by up to $1.57\times$ while avoiding expensive GPU runs during tuning. In collaboration with Lawrence Livermore National Laboratory (LLNL).
 - Profiled I/O bottlenecks and extended the Python framework for two large-scale weather forecasting applications.
- **HORATIO (SSDBM'26):** Develops an out-of-place index for gzip-compressed JSONL traces emitted by scientific applications, enabling fast queries directly over compressed data without format conversion and with minimal additional storage. In collaboration with LLNL.
 - Evaluated the native C++ analysis pipeline against raw and indexed JSON, and against other file formats such as Parquet and DuckDB, demonstrating up to $83\times$ faster analysis.

OPEN-SOURCE CONTRIBUTIONS

Google Summer of Code (GSoC)

Sep 2023 – Mar 2025

Open Source Research Experience (OSRE), UC Santa Cruz OSPO

Remote

- Built an extensible benchmark for concept drift in ML-for-systems workloads, comparing three online adaptation algorithms against retraining across four datasets with mentors from Argonne National Laboratory.
- Integrated Matchmaker, DriftSurf, and AUE into ML-based caching and storage systems, including GL-Cache and LinnOS, reducing cache misses by up to 5% while evaluating accuracy, training cost, and latency tradeoffs.
- Released open-source implementations, experiment results, and a reproducible Chameleon Cloud artifact.

SEACrowd

Jul 2023 – Sep 2024

Contributor and Community Reviewer

Bandung, Indonesia

- Implemented Hugging Face dataset loaders for under-resourced Southeast Asian languages as part of the SEACrowd open-data and benchmark ecosystem published at EMNLP'24.
- Reviewed 50+ community contributions for loader correctness, dataset metadata, and interface compatibility.

INDUSTRY EXPERIENCE

Bukit Vista Hospitality Services

Mar 2022 – Dec 2022

Backend Developer

Remote

- Built the backend for an in-house multi-property availability calendar, contributing to API design, database modeling, implementation, and deployment on AWS; enabled property owners to view booking availability and occupancy across their portfolios in one place.
- Automated finance and HR workflows previously handled manually, reducing processing time by up to 90%.
- **Technologies:** Node.js, Python, AWS.

TEACHING EXPERIENCE

CMSC 23000: Operating Systems University of Chicago

Aut 2025, 2026

CMSC 33200: Topics in Operating Systems University of Chicago

Spr 2026

CMSC 14400: Systems Programming II University of Chicago

Win 2026

TECHNICAL SKILLS

Languages: C, C++, Python, Go, Java, JavaScript, SQL, Bash/Shell

ML & Parallel Computing: PyTorch, TensorFlow, CUDA, MPI, OpenMP, SLURM

Systems & Data: CacheLib, Redis, Apache Spark, Pandas, PyArrow/Parquet, RabbitMQ, PostgreSQL

Infrastructure & Cloud: Linux, Git, Docker, Kubernetes, AWS, Google Cloud, Chameleon Cloud